

Design Model

What is the Design Model?

The **Design Model** is like a **blueprint** for your software.

It shows **how the system will work internally** and **how it will look and run** when finished.

It connects:

- **What users want (Requirements)** →
- **To How developers will build it (Design).**

The **Design Model** usually has **5 key elements**.

The 5 Design Model Elements

1. Data Design Elements

What it means:

This part decides **how your app will store, manage, and organize data**.

Examples:

- What data is needed? (e.g., User info, step count, workout records)
- What database will you use? (e.g., SQLite, Firebase)
- How will data flow? (e.g., Tables, schemas, ER diagrams)

Real-life:

It's like deciding where you'll keep your files in folders, with labels, so you can find them fast.

2. Architectural Design Elements

What it means:

This part shows the **big structure** the **main modules** and **how they connect**.

Examples:

- What are the main parts? (e.g., UI module, database module, API module)
- How do they talk to each other?
- Are you using layers? (like 3-tier architecture: Presentation, Business Logic, Data)

Real-life:

It's like planning the **floor plan of a house** — where are the rooms, doors, and pathways.

3. Interface Design Elements

What it means:

This part defines **how users and other systems interact** with your software.

It includes two main interfaces:**1. User Interface (UI):**

- Screens, buttons, forms.
- How users click, enter data, get feedback.

2. System Interface (APIs):

- How your software talks to other software.
- Example: Your app calls Google Maps API to find location.

Real-life:

It's like designing the **doors, windows, and switches** people use to interact with the house.

4. Component-Level Design

What it means:

This is the **detailed design** for each **small part or module**.

Decide:

- What functions/classes each module will have.
- What inputs/outputs they use.
- How they work inside.

Example:

- The Step Counter Service → How it gets sensor data, filters noise, saves steps.
- The Water Reminder → How it sets alarms, shows notifications.

Real-life:

It's like planning the **inside layout** of each room in your house furniture, wiring, plumbing.

5. Deployment-Level Design

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What it means:

This shows **how the software will run** on real hardware.

Decide decide:

- What devices will run the app? (Android phones, web browsers, servers)
- How is it installed? (APK, web server, cloud)
- How parts are distributed if it's a big system.

Example:

- FitTrack's database is stored locally on the phone.
- The step counting runs in the background.
- Notifications run through Android's OS.

Real-life:

It's like deciding where you'll **place the house** city, wiring to power grid, water connection.

Summary Table

Element	What it Covers	Easy Example
Data Design	How data is stored & structured	Tables, ER diagram, database schema
Architectural Design	Big structure & module connections	Floor plan: UI, DB, APIs
Interface Design	How users & systems interact	Screens, buttons, APIs
Component-Level Design	Internal design of each module	Step Counter class, Notification class
Deployment Design	Where & how software runs	Android phones, servers, cloud